Formula Sheet and Key Numbers

Print double-sided and keep it in your observing kit · hand out each unit's section when that unit begins

Unit 1 — Motion of the sky and time-keeping

Sky's rotation	$15^\circ/\text{h} = 1^\circ$ per 4 min $= 15'/\text{min} = 15''/\text{s}$
Sidereal day	23 h 56 m 4 s ⇒ stars rise ~4 min earlier each night (2 h per month)
Speed of Earth's surface	$v \approx 1670 \cos\varphi$ km/h
Axial tilt (obliquity)	$\varepsilon = 23.44^\circ$ tropical year 365.2422 d, sidereal year 365.2564 d
Noon altitude of the Sun	$h = 90^\circ - \varphi + \delta$ (if $> 90^\circ$: north of overhead, $180^\circ - h$)
Zero Shadow Day	$\delta_\odot = \varphi$ (only between 23.44° N and S)
Precession	1 cycle ≈ 25,800 yr; ≈ 50″/yr ≈ 1° per 72 yr; ayanamsha ≈ 24°
Moon	synodic 29.53 d, sidereal 27.32 d; rises ~50 min later daily; orbit tilt 5.1° (nodes Rahu & Ketu)
Saros · Foucault pendulum	18 y 11 d 8 h · $T = 23.93\ \text{h}/\sin\varphi$
Tithi	$\lceil E/12^\circ \rceil$ (E = Moon–Sun elongation); 1–15 Shukla, 16–30 Krishna
Nakshatra · rashi	$13^\circ 20'$ (27) · 30° (12)
Weekday order	Saturn, Jupiter, Mars, Sun, Venus, Mercury, Moon — jump 3 places a day ($24 = 3 \times 7 + 3$)
Adhik Maas	$12 \times 29.53 = 354.4$ d (10.9 d short); 7 extra months in 19 years (≈ every 32½ months)
Indian units	prana 4 s · pala 24 s · ghati 24 min · muhurta 48 min · prahar 3 h · day = 60 ghati
Eras	VS = CE + 57 · Shaka = CE − 78 · National Calendar: 1 Chaitra = 22 March (21 in leap years)
Hijri	354/355 d, no leap month, ~11 d earlier each year; AH ≈ (CE − 622) × $\frac{33}{32}$, CE ≈ AH × $\frac{32}{33}$ + 622
Gregorian leap year	divisible by 4, except centuries unless divisible by 400 (mean 365.2425 d)
IST and local time	IST = UTC + 5:30 (82.5° E); LMT = IST + 4 min×(λ − 82.5°); equation of time −14 to +16 min

Unit 2 — Sky, stars and constellations

Altitude of the celestial pole	= latitude φ
Transit altitude · lower culmination	$h = 90^\circ - \varphi + \delta \cdot h_{\text{low}} = \varphi + \delta - 90^\circ$
Circumpolar · never rises · overhead	$\delta > 90^\circ - \varphi \cdot \delta < -(90^\circ - \varphi) \cdot \delta = \varphi$
RA ↔ degrees	$1^\text{h} = 15^\circ$, $1^\text{m} = 15'$, $1^\text{s} = 15''$
Hour angle · on meridian	HA = LST − RA · LST = RA
Rough LST	LST ≈ $\alpha_\odot$ + (local time − 12^h); $\alpha_\odot \approx 0^\text{h}$ on 21 March, $+2^\text{h}$ per month
Small-angle formula	$\theta(^\circ) = 57.3\, D/d$
Hand at arm's length	finger 1° · three fingers 5° · fist 10° · spread hand 20°–25°
Handy sizes	Moon/Sun 0.5° · Orion's Belt ~ 3° · Dubhe–Merak 5.4° · Big Dipper ~ 25°

© Vaibhav Yenare

P.T.O.

Unit 3 — Magnitudes, telescopes and catalogues

Key steps	1 mag = 2.512× · 2.5 mag = 10× · 5 mag = 100× · 10 mag = 10^4×
Inverse square	$F \propto 1/d^2$; 2× farther = +1.5 mag; 10× farther = +5 mag · two equal stars combined: $m - 0.75$
Sun · distances	$m = -26.74$, $M = +4.83$ · 1 AU = 1.496×10^8 km · 1 pc = 3.26 ly = 206,265 AU
Focal ratio · magnification	$N = F/D$ · $M = F/f$ (× Barlow) · eyepiece for a target M: $f = F/M$
Useful magnification	$M_{\max} \approx 2D$ (mm) · $M_{\min} \approx D/7$
Exit pupil · true field	$D/M = f/N$ · TFOV ≈ AFOV/M
Light grasp	$(D/7\text{ mm})^2$ vs the eye · $(D_1/D_2)^2$ between telescopes
Resolution	Dawes $116/D$ arcsec · Rayleigh $138/D$ arcsec (D in mm)
Limiting magnitude	$m_{\lim} \approx 2.7 + 5\log_{10} D$ (mm) or $6 + 5\log_{10}(D/7)$
Imaging	size = $F\theta$ (rad) · plate scale $206265/F$ arcsec/mm · pixel scale $206.265\,p(\mu\text{m})/F$ arcsec/px

Magnitude and distance formulas:

$$\frac{F_1}{F_2} = 10^{0.4(m_2-m_1)} \qquad m_1 - m_2 = -2.5\log_{10}\frac{F_1}{F_2} \qquad M = m - 5\log_{10}\frac{d}{10\text{ pc}}$$

$$d = 10^{(m-M+5)/5}\text{ pc} \qquad \frac{L}{L_\odot} = 10^{0.4(4.83-M)}$$

Logarithm help: $\log_{10}$ of 1.5, 2, 2.5, 3, 4, 5, 6, 7, 8, 9 = 0.176, 0.301, 0.398, 0.477, 0.602, 0.699, 0.778, 0.845, 0.903, 0.954; $\log_{10}(ab) = \log_{10} a + \log_{10} b$, e.g. $\log_{10} 200 = 2.301$.